# tdm-dash · sensor module (Pico) for the Yamaha TDM 850 (3VD)

Schematic generated by hardware/kicad/generate.py. The wire colors and the pinout of the original connector are [TO VERIFY]: see docs/research/original-wiring.md.
Manual chapter 04: explanation of every sheet and component calculations.

1 Power

File: power.kicad_sch

2 Warning lights

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3 Tachometer

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5 Analog inputs

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6 MCU

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7 Backlight and Pi

File: backlight.kicad_sch

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8 sheets
**tdm-dash · CERN-OHL-P-2.0**

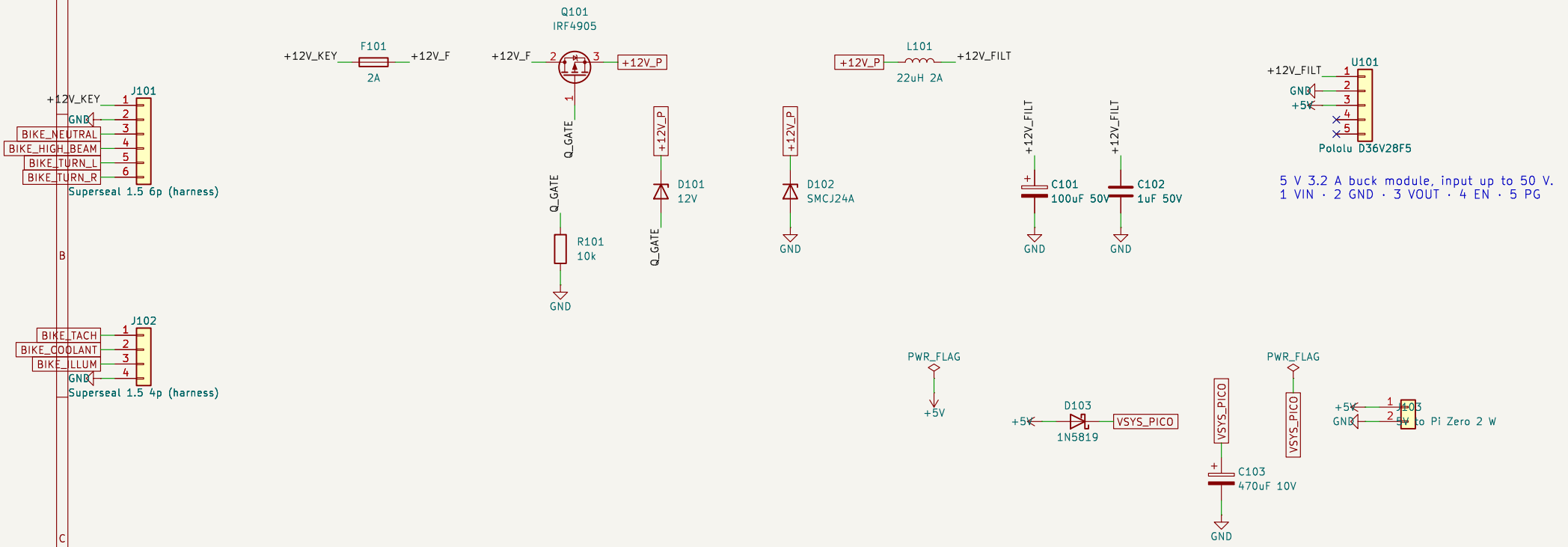
Sheet: /
File: tdm-dash.kicad_sch

***Title: tdm-dash · sensor module***

| Size: A4 | Date: 2026-10-21 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 1/9 |

# Power supply and protection

+12V_KEY: switched +12 V from the bike harness (Br wire [TO VERIFY]).
Chain: 2 A fuse → reverse-polarity P-MOSFET → TVS → LC filter → 5 V buck.
The Pico is supplied through D103 with 470 µF of hold-up, to save the km when the key goes off.




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Sheet: /1 Power/
File: power.kicad_sch

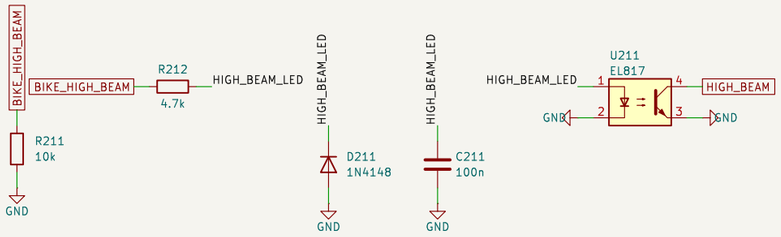
***Title: Power supply and protection***

| Size: A4 | Date: 2026-10-21 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 2/9 |

# Warning light inputs

High beam, turn signals and illumination are active at +12 V: 4.7 kΩ → EL817 LED → ground.
Neutral is active to ground (the gearbox switch grounds the wire):
+12V_P → 3.3 kΩ → LED → neutral wire. Outputs active low with the Pico's internal pull-up.
Wire colors and neutral logic [TO VERIFY] with the bike measurements.

## High beam

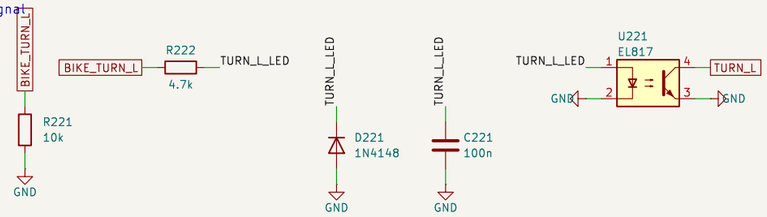






Outputs to the Pico
NEUTRAL → GP6
HIGH_BEAM → GP7
TURN_L → GP8
TURN_R → GP9
ILLUM → GP10

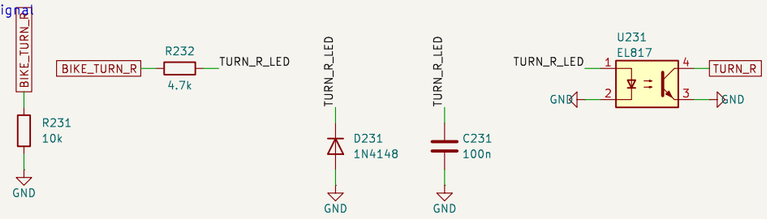
## Left turn signal







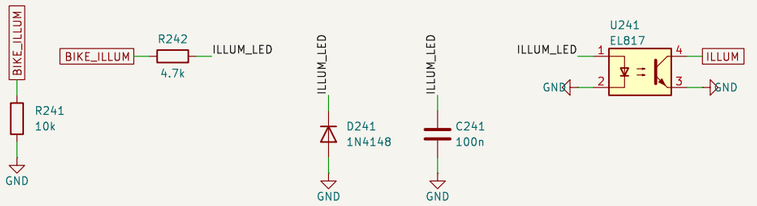
## Right turn signal







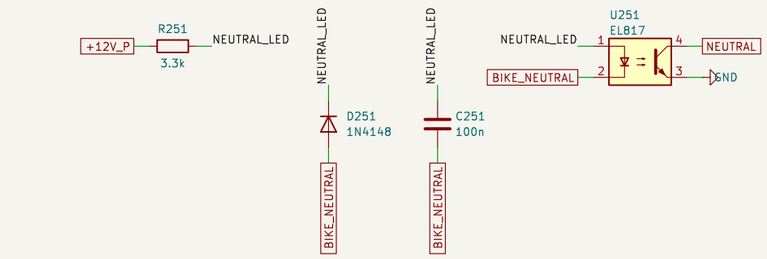
## Illumination







## Neutral (active to ground)







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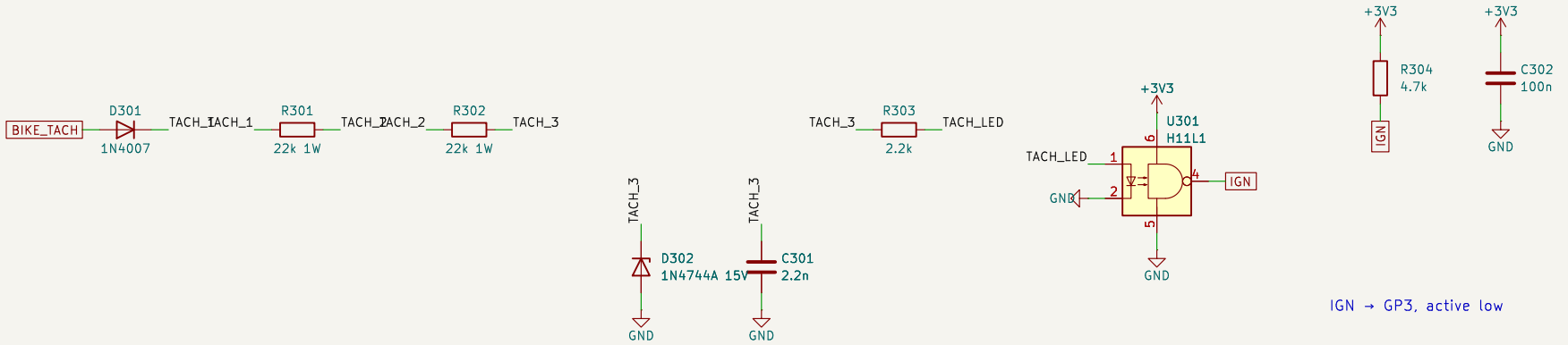
Sheet: /2 Warning lights/
File: inputs.kicad_sch

***Title: Warning light inputs (optocouplers)***

| Size: A3 | Date: 2026-10-21 | **Rev: 0.1** |
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# Tachometer

The coil negative has spikes of a few hundred volts at every spark.
1N4007 passes only the positive ones, 2 × 22 kΩ 1 W limit the current, the 15 V
zener and the 2.2 nF clip the spike; the H11L1 (Schmitt trigger) gives a clean edge.
Tachometer wire and waveform [TO VERIFY] with the oscilloscope.

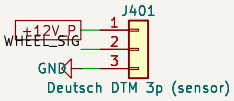


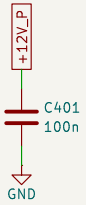

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| | | |
|---|---|---|
| Sheet: /3 Tachometer/ | | |
| File: tach.kicad_sch | | |
| ***Title: Tachometer (coil negative)*** | | |
| Size: A4 | Date: 2026-10-21 | **Rev: 0.1** |
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# Speed sensor

LJ12A3-4-Z/BX inductive NPN normally open: in front of a bolt the output goes to ground.
+12V_P → 2.2 kΩ → EL817 LED → sensor output. External 4.7 kΩ pull-up for fast edges.
Sensor colors: brown +12 V, blue ground, black signal.

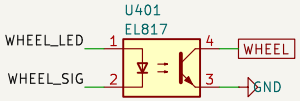


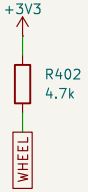










WHEEL → GP2, active low

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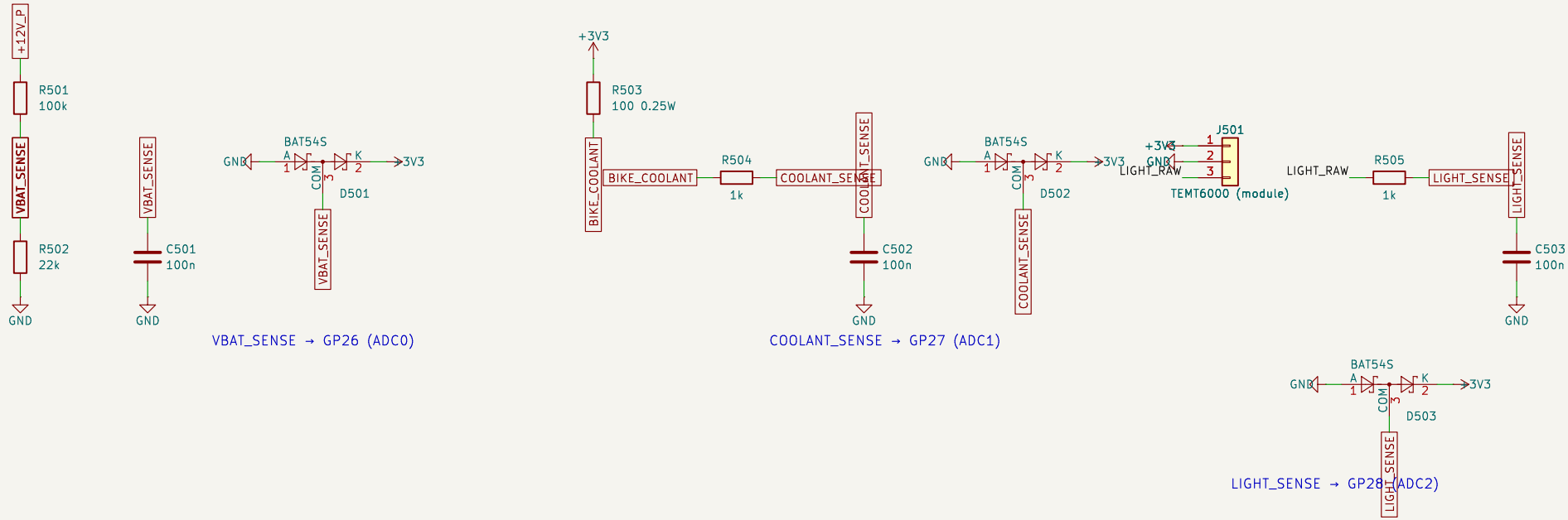
Sheet: /4 Wheel sensor/
File: wheel.kicad_sch

***Title: Wheel sensor (LJ12A3)***

| Size: A4 | Date: 2026-10-21 | **Rev: 0.1** |
|---|---|---|
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# Analog inputs

Battery: 100k/22k divider (18.3 V → 3.3 V), 100 nF filter, BAT54S clamp to 0 and 3.3 V.
Coolant: the NTC sender goes to ground in the engine; 100 Ω pull-up to 3.3 V, then 1 kΩ and clamp.
Light: TEMT6000 module supplied at 3.3 V.




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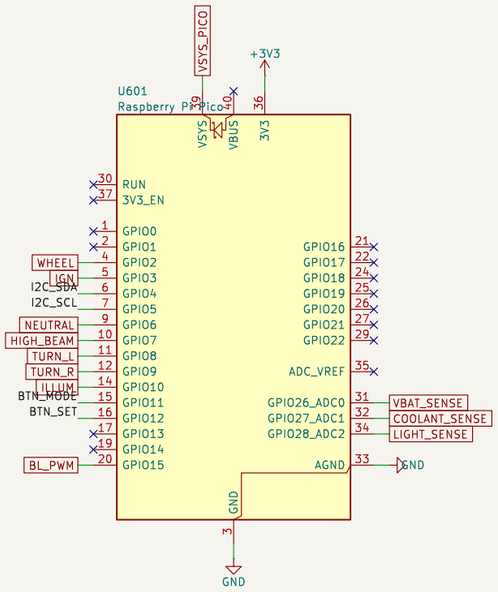
Sheet: /5 Analog inputs/
File: analog.kicad_sch

***Title: Analog inputs***

| Size: A4 | Date: 2026-10-21 | **Rev: 0.1** |
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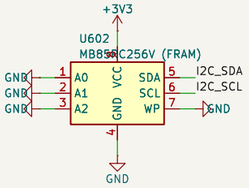
# Microcontroller, memory and buttons

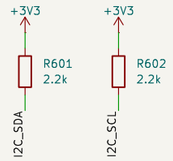
Raspberry Pi Pico (or Pico 2): same pinout, same firmware.
FRAM MB85RC256V at address 0x50 (A0-A2 to ground), I²C at 1 MHz.
MODE and SET buttons to ground, 1 kΩ + 100 nF against discharges and bounce.



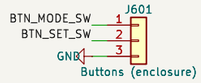


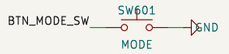




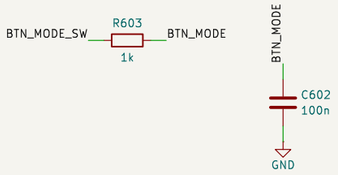

FRAM on a breakout (Adafruit 1895):
its header carries VCC, GND, SDA, SCL.

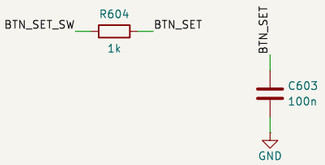









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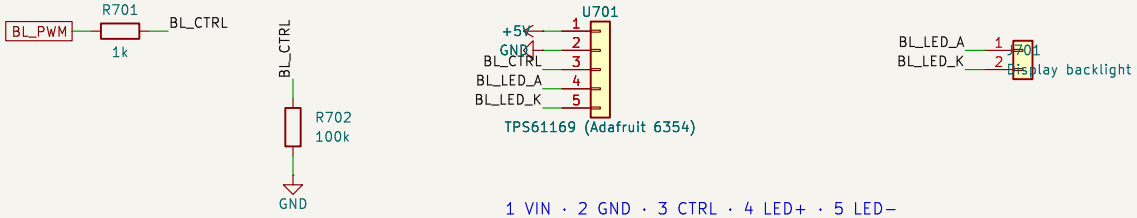
Sheet: /6 MCU/
File: mcu.kicad_sch

***Title: Pico, FRAM and buttons***

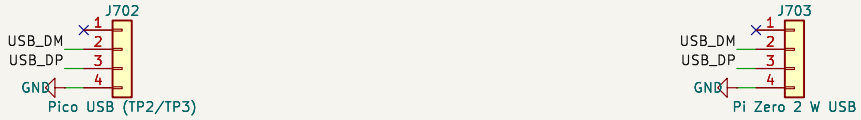
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| Size: A3 | Date: 2026-10-21 | **Rev: 0.1** |
| KiCad E.D.A. 9.0.9 | | Id: 7/9 |

# Backlight and USB to the Pi

The Pico drives the TPS61169 with 25 kHz PWM (GP15). 100 kΩ to ground: until the firmware runs, the backlight stays off (fail-dark).
Newhaven display: LED 180 mA at 9.3 V. Module and display pins [TO VERIFY].
USB: only D+, D− and ground. VBUS not connected: the Pico is supplied from VSYS.




1 VIN · 2 GND · 3 CTRL · 4 LED+ · 5 LED−




1 VBUS (not connected) · 2 D− · 3 D+ · 4 GND
Short twisted pair, ferrite on the cable.

The display connects to the Pi through the Adafruit DPI Kippah (40-pin FFC connector, off this board).

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Sheet: /7 Backlight and Pi/
File: backlight.kicad_sch

***Title: Backlight and link to the Pi***

| Size: A4 | Date: 2026-10-21 | **Rev: 0.1** |
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| KiCad E.D.A. 9.0.9 | | Id: 8/9 |

# Backup LEDs

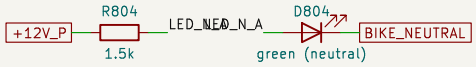
Wired straight to the bike's 12 V lines: they light up even with the electronics off. The turn signals use one pair of wires like the original indicator (Ch-Dg): the flashing side feeds its LED, the other line goes to ground through the bulbs. The 1N4148 in series protects each LED from the reverse 12 V.










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Sheet: /8 Backup LEDs/

File: leds.kicad_sch

***Title: Backup LEDs***

| Size: A4 | Date: 2026-10-21 | **Rev: 0.1** |
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| KiCad E.D.A. 9.0.9 | | Id: 9/9 |